

Case and agreement in Kalaallisut non-finite constructions

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1 Introduction

Kalaallisut (Inuit, Greenland) has a small inventory of negative verbs that occur with a non-finite form of a lexical verb to form a periphrastic negative construction (PNC). The non-finite form of the lexical verb is formed with *-neq* and the most common negative verbs are *ajor-* (habitual negation, lit. ‘be bad’), *ajornar-* ‘be impossible’, *artor-* ‘be unable’, *nalu-* ‘not know’, and *saper-* ‘not manage’. These are illustrated in (1)–(5). We focus on *ajor-* throughout the paper, but briefly discuss *ajornar-*, *artor-*, *nalu-* and *saper-* in section 7.^{1,2}

- (1) Qatannguta–asa qajassuar–un–neq **ajor**–paat.
sibling–3SG>3PL.ERG be.gentle–APPL–NEQ HAB.NEG–3PL>3SG.IND
‘His siblings didn’t spare him any more.’ Paakujuk (Sooty), §9, ex 7
- (2) Sinin–neq **ajorna**–qa–aq
sleep–NEQ be.impossible–more.and.more–3SG.IND
‘It’s really impossible to sleep.’ Fortescue (1984:179)
- (3) Naja–p karsi kivin–neq **artor**–paa.
Naja–ERG box.ABS lift–NEQ be.unable–3SG>3SG.IND
‘Naja can’t lift the box.’ Grethe Schmidt BPI-GS-2
- (4) Naja mersor–neq **nalu**–voq.
Naja sew–NEQ not.know–3SG.IND
‘Naja doesn’t know how to sew.’ Ellen Thrane CIT-3

¹Elicited and volunteered examples are annotated with the name of the speaker and an alpha-numerical code identifying it in the second author’s notebooks. Textual examples primarily come from a collection of Kalaallisut texts transcribed and glossed by Maria Bittner (<https://sites.google.com/view/maria-bittner/kalaallisut>). These are annotated with the name of the story, paragraph and example number. A few of the textual examples in the appendix come from personal narratives by the third and fourth author. Finally, examples from the literature are adapted to current standard orthography and glossed in accordance with footnote 2.

²Glossing conventions: ABL = ablative case; ABS = absolutive case; ALL = allative case; AP = antipassive; APPL = applicative; CAU = causative mood; CAUS = causativizer; CON = conditional mood; CONJ = conjunction; COP = copula; EQUA = equalis case; ERG = ergative case; FOC = focus; FRUST = frustrative; FUT = future; HAB = habitual; IND = indicative mood; INST = instrumental; INT = interrogative mood; LOC = locative case; NEG = negative; NOMZ = nominalizer; PART = participial mood; PASS = passive; PL = plural; SG = singular. 3SG>3SG.IND means third person singular subject acting on third person singular object in indicative mood. 1SG>3SG.ERG means 1SG possessor with 3SG possessum in the ergative case. As for *-neq*, Bittner (1994) glosses it GER for ‘gerund’, whereas Fortescue (1984) and Kahn and Valijärvi (2022) gloss it AP for ‘abstract particle’. Since the term gerund implies nominal and we do not take *-neq* to have a nominalizing function, we do not adopt Bittner’s gloss here. We also do not use the abstract participle AP gloss, partly because we have reserved AP for ‘antipassive’, partly because the meaning of this term is somewhat obscure to us. Instead we simply gloss *-neq* as NEQ, and defer the discussion of its morpho-syntactic status and function to section 5. Finally note that clauses unmarked for tense can receive either a present or past interpretation, including in the PNC.

- (5) oqarfigi–neq **saper**–para.
 say.to–NEQ not.manage–1SG>3SG.IND
 ‘I cannot/dare not talk to him (about it).’

Fortescue (1984:48)

The PNC is noted in some descriptive grammars (Fortescue 1984:47–48, Kahn and Valijärvi 2022:245–247, and Sadock 2003:10–11, 33) and in Woodbury (1975:70–71), but, to the best of our knowledge, no systematic description or analysis has been developed. We aim to fill this gap here. We argue that the negative verbs are negative auxiliaries that take a non-finite VP as their complement (in accord with Sadock 2003:10). The PNC is superficially similar to control structures, which also feature a lexical verb affixed with *-neq* below a finite verb. Compare (6) to (4).

- (6) Naja–p mensor–neq nuannari–vaa.
 Naja–ERG sew–NEQ love–3SG>3SG.IND
 ‘Naja loves to sew.’

Ellen Thrane CHV-1

We argue that in (6) the *neq*-phrase is nominalized and serves as a DP complement to the control verb *nuannari*- ‘love to’ (see Mourounas 2019 for detailed discussion of control into nominalizations).³

By contrasting the PNC with control constructions we can affirm and refine an existing generalization about the distribution of absolutive case in the language. Specifically, our investigation affirms that there is at most one absolutive argument within a case domain and that the relevant case domain is the *finite* clause. The existence of negative auxiliaries in Kalaallisut has not been noted in existing typologies of negation (Dahl 1979, WALS Feature 112A) and our study thus adds a data point to a somewhat rare negation strategy (only 4% of the 1157 WALS language sample utilize negative auxiliaries). Finally, the existence of a periphrastic negation construction within a deeply polysynthetic language is interesting in and of itself. As expected, regular negation is affixal (*-nngit*) but in addition the language has this multi-word construction for habitual negation. From a theoretical perspective, the PNC is also puzzling. In their influential paper on polysynthesis in Inuit, Compton and Pittman (2010) argue that only phrases form words and only CP and DP are phrases. If we are correct that the complement of negative auxiliaries are non-finite VPs, this strong hypothesis must be modified.

The paper is structured as follows. In section 2, we provide background on our collaboration, including data collection, as well as ecological and relevant linguistic properties of Kalaallisut. In Section 3 we show that the *neq*-forms of the lexical verb that are found in both the PNC and control structures live a double life: in the PNC the *neq*-phrase is purely verbal; in control constructions it is nominalized. In section 4 we examine the predictions that flow from this contrast and show that they are borne out. In section 5 we develop syntactic structures for the PNC and control structures that capture their differing behavior and situate the negative auxiliaries in the verbal projection. Section 6 draws out the implications for case assignment in the language and in section 7 we briefly discuss the other negative auxiliaries and show that some of them participate in both the PNC construction and in control. Section 8 concludes.

³We have preliminary evidence for a second control construction in which the *neq*-phrase appears to not be nominalized and the control verb can agree with the internal argument of the embedded verb. This construction appears to have gone under the radar in existing discussions of control. Bittner (1994:53–54, 64–66) and Fortescue (1984:44–48) take all *neq*-derivations to be nominalizations and we have not been able to find any discussion of control structures in Sadock (2003). We leave investigation of this second control construction for future research.

2 Background

This work stems from a collaboration between linguists Line Mikkelsen & Tyler Lee-Wynant and Kalaallisut speakers Grethe Schmidt & Ellen Thrane.

Ellen Thrane’s autobiography I was born in Upernavik in north west Greenland. My mother and grandmother spoke only Kalaallisut, whereas my father spoke Kalaallisut and Danish. I acquired Kalaallisut from birth and only started learning Danish in school. I moved south to Ilulissat at age 9 and to Denmark after 7th grade for educational opportunities. I volunteer as a Kalaallisut-Danish-Kalaallisut interpreter for Danish public agencies and am affiliated with Umiaq, a Greenlandic association hosted at Kalaallit Illuutaat (‘The Greenlanders’ house’) in Odense, Denmark. Kalaallisut is my mother tongue and identity.

Grethe Schmidt’s autobiography I was born in Attu in north west Greenland and spoke Kalaallisut at home and in school. I lived in a number of other places in Greenland before moving to Denmark in 1980. I am a Kalaallisut-Danish-Kalaallisut translator for Danish TV-channels and also affiliated with Umiaq at Kalaallit Illuutaat. Kalaallisut is the language I am born with.

Line Mikkelsen and Ellen Thrane began collaborating in summer of 2022, Grethe Schmidt joined the project in spring of 2024 and Tyler Lee-Wynant in Fall of 2024. The data discussed below come from one-on-one weekly Zoom elicitation sessions, which also included discussion of linguistic questions of interest. Elicitation was conducted by Line Mikkelsen in Danish. Danish prompts and translations were subsequently translated into English by the second author. All elicitation recordings are archived in the California Language Archive. With the exception of the scrambling data in (31), judgments were overall uniform, within speaker and between speakers. Key data points were additionally checked with native-speaker linguist Naja Blytman Trondhjem.

Language ecology Kalaallisut is a member of the Inuit-Yupik-Unangan language family, which is spoken from Eastern Russia across Alaska and Canada to Greenland. It is the sole official language of Greenland, with around 50,000 speakers in Greenland and another 17,000 in Denmark (Naja Blytman Trondhjem, p.c.). The language is acquired by children in the home and taught in school. Danish is/was a colonial power in Greenland and Danish language has a big presence, especially in administration. English has a growing presence.

Three linguistic aspects of Kalaallisut are of immediate relevance for our study: i) polysynthesis, ii) case and iii) verbal agreement.

Polysynthesis Kalaallisut manifests classic features of polysynthesis. First, verbs may exhibit a high morpheme-to-word ratio, as in the textual examples in (7) and (9).

- (7) Qulingilua-a-su-ssa-a-galuar-put.
nine-COP-3SG.PART-FUT-COP-FRUST-3PL.IND
‘There should’ve been nine.’ *Paakujuk* (Sooty), §10, ex 11
- (8) Nukappiaqqa-p unit-sin-niar-aluar-pai.
boy-ERG.SG stop-CAUS-try-FRUST-3SG>3PL.IND
‘The boy tried to stop them [= the dogs].’ *Paakujuk* (Sooty), §27, ex 2

- (9) ... navia–nar–tor–sior–tip–paanga.
 fear–induce–NOMZ–experience–CAUS–3SG>1SG.IND
 ‘... it put me in danger’ [lit. it made me experience having fear induced]
Piniartup uqaluttuaa (A hunter’s story), §5, ex 4

As these examples also show, there is rich verbal inflectional morphology, specifically mood and subject and object agreement. These categories are expounded together in a single, obligatory word-final suffix. Third, Kalaallisut exhibits grammatically free argument drop and noun incorporation. This is illustrated in (10), where all DP arguments are null pronouns and the notional object of ‘consume’ is incorporated.

- (10) *pro* kaagi–sor–uit *pro* *pro* ajori–ssu–akkit.
 2SG.ABS cake–consume–2SG.CON 1SG.ERG 2SG.ABS angry–FUT–1SG>2SG.IND
 ‘If you eat cake, I’ll be angry with you.’ Ellen Thrane ANM-1

Finally, Kalaallisut has grammatically free word order with basic SOV order (Fortescue 1984:93ff; Fortescue 1993).

Case Case alignment is ergative-absolutive. In (11) the transitive subject is marked with the ergative suffix *-p*. In contrast the transitive object and intransitive subject (12) are absolutive, which is null. The case distinction is only morphologically expressed in names and singular count nouns. We nonetheless gloss all core arguments as ergative or absolutive in accord with their syntactic function.

- (11) Naja–**p** Juuna ikior–paa.
 Naja–ERG Juuna.ABS help–3SG>3SG.IND
 ‘Naja helped Juuna.’ Ellen Thrane AFX-1
- (12) Juuna angerla–jaar–poq.
 Juuna.ABS come.home–early–3SG.IND
 ‘Juuna came home early.’ Ellen Thrane AFX-2

There is at most one absolutive argument per clause (Compton 2018:164). Therefore, two internal arguments of a ditransitive verb cannot both be absolutive (13). Instead, one must be realized as an oblique (14).

- (13) *Meeqqa–t poortukka–t nassip–pakka.
 child–PL.ABS package–PL.ABS send–1SG>3PL.IND
 Intended: ‘I sent the children the packages.’ Ellen Thrane CKS-2
- (14) Meeqqa–t poortukka–nik nassip–pakka.
 child–PL.ABS package–PL.INST send–1SG>3PL.IND
 ‘I send the children the package.’ Ellen Thrane CKS-3

Agreement Verbs agree with all absolutive and ergative arguments. (15) is ungrammatical because the verb only agrees with the subject. (16) is ungrammatical because the verb only shows agreement with the object. The only grammatical realization of a transitive clause is as in (17) where the verb agrees with both subject and object.

- (15) *Naja-p meeqa-t ikior-poq
 Naja-ERG child-PL.ABS help-3SG.IND
 Intended: ‘Naja helped the children.’ Grethe Schmidt CIE-GS-1
- (16) *Naja-p meeqa-t ikior-put.
 Naja-ERG child-PL.ABS help-3PL.IND
 Intended: ‘Naja helped the children.’ Grethe Schmidt CIE-GS-2
- (17) Naja-p meeqa-t ikior-pai.
 Naja-ERG child-PL.ABS help-3SG>3PL.IND
 ‘Naja helped the children.’ Grethe Schmidt CIE-GS-3

Verbs do not agree with oblique arguments:

- (18) Ilin-nut allap-punga.
 2SG-ALL write-1SG.IND
 ‘I wrote to you.’ Ellen Thrane CHT-1
- (19) *Ilin-nut allap-pakkit.
 2SG-ALL write-1SG>2SG.IND
 Intended: ‘I wrote to you.’ Ellen Thrane CHT-2

In sum, verbs agree with all and only absolutive and ergative arguments.

3 The duality of *-neq*

In addition to the PNC (20) *neq*-marked verbs feature in control constructions (21).

- (20) Naja **biiler-neq** ajor-poq.
 Naja.ABS drive-NEQ HAB.NEG-3SG.IND
 ‘Naja doesn’t drive.’ Ellen Thrane BPA-3
- (21) Naja-p **biiler-neq** nuannari-vaa.
 Naja-ERG drive-NEQ love-3SG>3SG.IND
 ‘Naja loves to drive.’ Ellen Thrane CHB-4

Despite their surface similarity, we argue that the status of the *neq*-phrase differs. In (20) *biiler-neq* is a non-finite verbal projection. In (21) that non-finite verbal projection is further nominalized.⁴ There are four pieces of evidence that the *neq*-phrase is nominal in control structures but not in the PNC.

⁴This duality of V-*neq* is reminiscent of the duality of English *-ing* forms, which also have verbal (participial) and nominal (gerund) guises (Abney 1987:167–244). This possible parallel deserves to be investigated but we will not attempt to do so here.

First, in control structures, the *neq*-phrase can bear possessive morphology (22) and (23); in the PNC it cannot, whether or not *ajor*- bears intransitive (24) or transitive (25) agreement.

- (22) Naja-p mersor-ni-**ni** nuannari-vaa.
 Naja-ERG sew-NEQ-4SG>3SG.ABS love-3SG>3SG.IND
 ‘Naja_i loves her_i sewing.’ Ellen Thrane CHR-1
- (23) sumiin-ni-**ni** paasi-nngil-aa.
 be.where-NEQ-4SG>3SG.ABS understand-NEG-3SG>3SG.IND
 ‘He didn’t understand where he was.’ Fortescue (1984:43)
- (24) *Naja mersor-ni-**ni** ajor-poq.
 Naja.ABS sew-NEQ-4SG>3SG.ABS HAB.NEG-3SG.IND
 Intended: ‘Naja doesn’t sew.’ Ellen Thrane CHR-2
- (25) *Naja-p mersor-ni-**ni** ajor-paa.
 Naja-ERG sew-NEQ-4SG>3SG.ABS HAB.NEG-3SG>3SG.IND
 Intended: Naja doesn’t sew. Ellen Thrane CII-7

Second, when embedded under a control verb, V-*neq* can host the nominal future tense morpheme -*ssaq* (26). This is not possible in the PNC (27). (On the nominal character of -*ssaq* see Fortescue 1984:218.)

- (26) Naja-p mersor-ni-ssa-ni nuannari-vaa.
 Naja-ERG sew-NEQ-FUT-4SG>3SG.ABS love-3SG>3SG.IND
 ‘Naja is looking forward to sewing.’ Grethe Schmidt CKN-GS-1
- (27) *Naja-p mersor-ni-ssa-ni ajor-paa.
 Naja-ERG sew-NEQ-FUT-4SG>3SG.ABS HAB.NEG-3SG>3SG.IND
 Intended: ‘Naja is not going to sew.’ Grethe Schmidt CKP-GS-1

Third, control verbs like *nuannari*- ‘love’ can take underived nominal complements (28), the negative auxiliary *ajor*- cannot (29):

- (28) Sialuk nuannari-vara.
 rain.ABS love-1SG>3SG.IND
 ‘I love rain.’ Grethe Schmidt CKJ-GS-2
- (29) *Sialuk ajor-poq.
 rain.ABS HAB.NEG-3SG.IND
 Intended: ‘It doesn’t rain.’ Grethe Schmidt CKJ-GS-1

These differences in possessive and tense morphology and subcategorization fall into place if V-*neq* is verbal in the PNC but nominal in control structures.

A fourth potential contrast between control constructions and the PNC comes from the claim in Sadock (2003:10, 33) that in the PNC, the *neq*-marked verb must be immediately adjacent to the negative auxiliary. This is clearly not the case for control verbs. Examples like (30), where the *neq*-phrase has scrambled over the subject, are always judged grammatical.

- (30) Mersor-neq Naja-p nuannari-vaa.
 sew-NEQ Naja-ERG love-3SG>3SG.IND
 ‘Naja loves to sew.’

Grethe Schmidt CIC-GS-2

This word order difference would follow from the *neq*-phrase being a DP in control structures but a VP in the PNC, since DPs can scramble but VP cannot (Grewendorf and Sternefeld 1990:13–14). However, in our work, scrambling of the *neq*-phrase in a PNC receives mixed judgments.

- (31) ILLUSTRATIVE EXAMPLE; NO JUDGMENT INCLUDED

Mersor-neq Naja ajor-poq.
 sew-NEQ Naja.ABS HAB.NEG-3SG.IND

Intended: ‘Naja doesn’t sew.’

The third author rejected examples like (31) on four out of six occasions and the fourth author rejected it on three out of five occasions. One possible interpretation of this state of affairs is that the ban on scrambling of VP, which was established mostly based on languages with a very different typological profile, is weaker in Kalaallisut and manifests as VP scrambling being of variable acceptability. Under that interpretation, the acceptability contrast between (30) and (31) supports our claim that in control structures the *neq*-phrase is a DP, whereas it is verbal in the PNC.

Anticipating our analysis in section 5, we propose that *-neq* uniformly heads a non-finite verbal projection, which we label γP . This straightforwardly explains its behavior in the PNC. In control structures, there is covert nominalization of the γP , which allows the embedded verb to express nominal morphology and the entire γP to scramble. This analysis is also consistent with control verbs, but not negative auxiliaries, being able to take underived nominals as their complement. Contra Fortescue (1984), Bittner (1994) and Sadock (2003:47), we thus do not take *-neq* to be a nominalizer, but instead a non-finite verbal category. Nominalization is zero-derived and present only in control structures. See Iordăchioaia and Melloni (2023) for a recent overview of zero-derived nominalizations.

4 Case and agreement

The discussion above lets us posit the following schematic structures for the PNC (32) and control (33) :

- (32) Subj [_{VP} V-*neq*] Aux PNC
 (33) Subj [_{DP} V-*neq*] V CONTROL

Now we examine the predictions about case and agreement that follow from these structures together with general principles of case and agreement in the language. We start with control.

4.1 Control

Our starting point for control structures is that the *neq*-phrase complement of the control verb is a DP. This makes five predictions about case and agreement which we consider in turn.

Prediction 1 As an internal DP argument of a transitive verb, the *neq*-phrase is absolutive. This claim is hard to assess in basic control constructions because absolutive case is null. However, possessive *neq*-DPs show distinctive absolutive morphology shared with possessed underived nominals (34) & (35). This absolutive possessive morphology contrasts with possessive morphology in other cases, including ergative possessive morphology in (36).

- (34) Hansi-p qimmi-**ni** nuannari-vaa.
Hansi-ERG dog-4SG>3SG.ABS love-3SG>3SG.IND
‘Hansi_i loves his_i dog.’ Grethe Schmidt CIH-GS-1
- (35) Naja-p mersor-ni-**ni** nuannari-vaa.
Naja-ERG sew-NEQ-4SG>3SG.ABS love-3SG>3SG.IND
‘Naja loves to sew./Naja loves her sewing’ Ellen Thrane CHR-1
- (36) Hansi-p qimmi-**ata** kii-vaanga.
Hansi-ERG dog-4SG>3SG.ERG bite-3SG>1SG.IND
‘Hansi’s dog bit me.’ Grethe Schmidt CIH-GS-2

Prediction 2 As the subject of a verb with an absolutive object, the subject is ergative (Yuan 2018). This holds true even if the embedded verb is intransitive, as seen in (37) and (38).

- (37) **Naja-p** biiler-neq nuannari-vaa.
Naja-ERG drive-NEQ love-3SG>3SG.IND
‘Naja loves to drive.’ Ellen Thrane CHB-4
- (38) ***Naja** biiler-neq nuannari-vaa.
Naja.ABS drive-NEQ love-3SG>3SG.IND
Intended: ‘Naja loves to drive.’ Ellen Thrane CHB-5

Prediction 3 Since there is at most one absolutive DP per CP (Compton 2018:164), it should not be possible to realize an internal argument of the *neq*-marked verb as an absolutive. This is indeed impossible:

- (39) *Meeqqa-nut poortukka-t nassi-un-neq nuannari-vara.
child-PL.ALL package-PL.ABS send-APPL-NEQ love-1SG>3SG.IND
Intended: I love to send packages to the children. Ellen Thrane CKI-3
- (40) *Iipili-t neri-neq qatsup-para.
apple-PL.ABS eat-NEQ be.tired.of-1SG>3SG.IND
Intended: ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-4

Instead such internal arguments are either incorporated (41) or realized as an oblique, e.g. an instrumental object of an antipassivized verb (42a) or an instrumental object of a notionally transitive, but formally intransitive verb form (42b); see also Fortescue (1984:45).

- (41) a. Puisi-tor-neq nuannara-arput.
seal-consume-NEQ love-1PL>3SG.IND
‘We love to eat seal meat.’ Ellen Thrane BSY-2

- b. *lipili-tor-neq qatsup-para.*
apple-consume-NEQ be.tired.of-1 SG>3SG.IND
 ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-6
- (42) a. *Meeqqa-nik ikiu-i-neq nuannari-vara.*
child-PL.INST help-AP-NEQ love-1 SG>3SG.IND
 ‘I love to help the children.’ Grethe Schmidt CHC-GS-2
- b. *lipili-nik neri-neq qatsup-para.*
apple-PL.INST eat-NEQ be.tired.of-1 SG>3SG.IND
 ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-5

Prediction 4 If a control construction makes an oblique case available, the *neq*-phrase should be able to expone that case. We see this in the example in (43), where the matrix verb ‘be tired’ licenses a source argument. That source argument is realized with ablative case on the *neq*-phrase. Because the *neq*-phrase is not absolutive, the internal argument of *malisor*- ‘chase’ can be realized as an independent absolutive DP without running afoul of the single absolutive restriction. Compare (43) to (39) and (40).

- (43) *Juuna-p qilalukka-t malisor-ner-mit qasu-voq.*
Juuna-ERG whale-PL.ABS chase-NEQ-ABL be.tired-3SG.IND
 ‘Juuna is tired from chasing whales.’ Bittner (1994:53), ex. 9a

Prediction 5 As an absolutive argument of a transitive verb, the *neq*-DP should control object agreement on the matrix control verb. As a nominalization, the ϕ -features of the *neq*-DP are 3SG (e.g. Hanink 2021, Picallo 2001). We thus correctly predict obligatory 3SG object agreement in control constructions.

- (44) *Mersor-neq nuannari-vara.*
sew-NEQ love-1 SG>3SG.IND
 ‘I love to sew.’ Ellen Thrane BTM-3
- (45) **Mersor-neq nuannari-vunga.*
sew-NEQ love-1 SG.IND
 Intended: ‘I love to sew.’ Ellen Thrane BTM-2

An alternative interpretation of (44) is that the 3SG object agreement is a form of default agreement. There are two reasons to resist that interpretation. First, coordinated *neq*-phrases can trigger 3PL object agreement:

- (46) *Erinarsor-neq qitin-ner-lu nuannari-vakka*
sing-NEQ dance-NEQ-CONJ love-1 SG>3PL.IND
 ‘I love to sing and dance.’ Ellen Thrane DJE-5

This is direct evidence for a genuine number feature on the *neq*-phrase and unexplained on a default agreement analysis. Second, we don’t see default object agreement in other contexts where person-number features of a notional object are inaccessible for agreement. Recall that verbs only agree

with ergative and absolutive arguments. There is a class of notionally transitive verbs, including *neri-* ‘eat’, which occur in two case frames: ERG-ABS and ABS-INST.⁵

In the ERG-ABS case frame, the verb shows transitive agreement:

- (47) Naja-p iipili-t pingasut neri-vai.
 Naja-ERG apple-PL.ABS three.ABS eat-3SG>3PL.IND
 ‘Naja ate three apples.’ Grethe Schmidt CKL-GS-1

Since verbs can’t agree with oblique arguments, we might expect default 3SG object agreement in the ABS-INST frame. But that is impossible (48). Instead we find intransitive agreement (49).

- (48) *Naja iipili-nik pingasu-nik neri-vaa.
 Naja.ABS apple-INST.PL three-INST.PL eat-3SG>3SG.IND
 Intended: Naja ate three apples. Grethe Schmidt CKL-GS-3
- (49) Naja iipili-nik pingasu-nik neri-voq.
 Naja.ABS apple-INST.PL three-INST.PL eat-3SG.IND
 Naja ate three apples. Grethe Schmidt CKL-GS-4

In other words, outside of control structures, there is no evidence for default agreement. In fact there is evidence against it from the ungrammaticality of (48).

To summarize: the hypothesis that the non-finite *neq*-phrase is encased in a null nominal shell captures the case and agreement patterns found in control constructions.

4.2 Periphrastic Negation structures

For the PNC, the starting point is that *V-neq* is a non-finite VP with no nominalizing layer of syntactic structure. From this hypothesis four predictions follow.

Prediction 1 As a VP, the *neq*-phrase cannot bear any case morphology:

- (50) *Naja-p mersor-ni-ni ajor-paa.
 Naja-ERG sew-AP-4SG>3SG.ABS HAB.NEG-3SG>3SG.IND
 Intended: Naja loves to sew. Ellen Thrane CII-7
- (51) *Naja biiler-nem-mik ajor-poq.
 Naja.ABS drive-NEQ-INST HAB.NEG-3SG.IND
 Intended: Naja doesn’t drive. Grethe Schmidt BRF-GS-4
- (52) *Meeqqa-t ikior-nim-mut ajor-pakka.
 child-PL.ABS help-NEQ-ALL HAB.NEG-1SG>3PL.IND
 Intended: I didn’t help the children. Grethe Schmidt BRG-GS-3

⁵ According to traditional treatments like those in Kleinschmidt (1851/1968), Bergsland (1955), Woodbury (1975), Fortescue (1984), and Sadock (1980), absolutive objects are definite and instrumental objects are indefinite. Bittner (1987) shows that the situation is considerably more complex and argues that the relevant difference is one of scope: absolutive objects take wide scope, whereas instrumental objects must take narrow scope. In our work there is not a consistent correlation between an absolutive case-marked object and a definite translation nor between an instrumental case-marked object and an indefinite translation. Since we are not concerned with semantic interpretation of the constructions under discussion, we set this issue aside and simply provide faithful English translations of the Danish translations established in elicitation.

Prediction 2 Without an absolutive internal argument of the lexical verb, the subject cannot be ergative, whether the negative auxiliary shows transitive (53) or intransitive (54) agreement:

- (53) *Naja-p biiler-neq ajor-paa.
 Naja-ERG drive-NEQ HAB.NEG-3SG>3SG.IND
 Intended: Naja doesn't drive. Grethe Schmidt CIH-GS-4
- (54) *Naja-p biiler-neq ajor-poq.
 Naja-ERG drive-NEQ HAB.NEG-3SG.IND
 Intended: Naja doesn't drive. Grethe Schmidt CIH-GS-3

Prediction 3 Since the *neq*-phrase is not absolutive, absolutive case is available for internal arguments of the lexical verb, as seen for *meeqqa-ni* 'her children' in (55).

- (55) Naja-p **meeqqa-ni** ikior-neq ajor-pai.
 Naja-ERG child-4SG>3SG.ABS help-NEQ HAB.NEG-3SG>3PL.IND
 'Naja_i didn't help her_i children.'
 Grethe Schmidt BRJ-GS-2

Prediction 4 As the *neq*-phrase is not absolutive, it cannot control object agreement on the negative verb:

- (56) *Biiler-neq ajor-para.
 drive-NEQ HAB.NEG-1SG>3SG.IND
 Intended: 'I don't drive.'
 Grethe Schmidt CHA-GS-2

To summarize: in the PNC the *neq*-phrase is syntactically inert as regards case and agreement. It does not bear case itself, it does not license ergative case on the higher subject, and it does not control agreement. This follows straightforwardly on the hypothesis that in the PNC the *neq*-phrase is exclusively verbal.

5 The structure of *neq*-phrases

Three questions arise at this point:

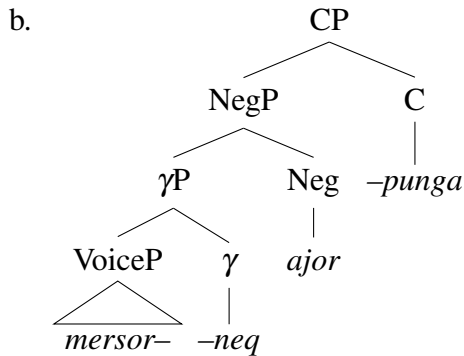
1. what is the syntactic category of *-neq*?
2. what is the structure below *-neq*?
3. what is the structure above *-neq*?

We will argue that: *-neq* instantiates a γ head. γ P can include verbal functional heads that govern argument structure (passive, antipassive, causative, applicative, and ditransitive alternations), but no functional heads associated with inflectional morphology (mood and agreement). In the PNC

there are no nominal functional heads above γP ; negative auxiliaries select directly for γP . In control structures γP is encased in a DP-shell and control verbs select for that DP.⁶ This analysis is represented in the tree diagrams below for the PNC (57) and control (58). For ease of exposition, we use VoiceP to represent the highest projection in the argument structure domain. As such it can embed projections headed by applicative, causative, and antipassive morphemes. Following Compton (2018), we take mood and agreement to be exponed in C.

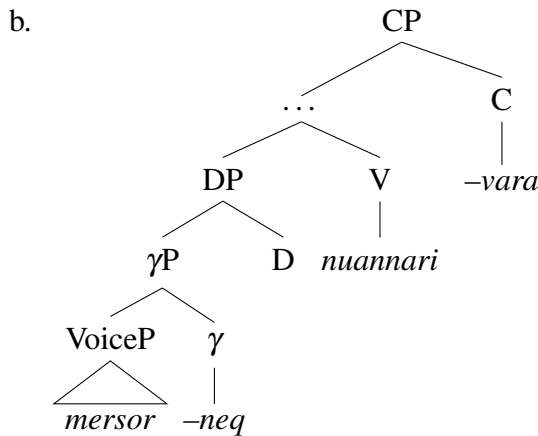
- (57) a. Mersor-neq ajor-punga.
 sew-NEQ HAB.NEG-1SG.IND
 ‘I don’t sew.’

Ellen Thrane CJP-4



- (58) a. Mersor-neq nuannari-vara.
 sew-NEQ love-1SG>3SG.IND
 ‘I love to sew.’

Grethe Schmidt BTR-GS-1



⁶According to Iordăchioaia (2020), only nominalization by n carries ϕ -features; nominalization directly by D do not. Since *neq*-phrases in control structures have genuine ϕ features (see section 4), this view would entail a n-layer between γP and D. The relevant contrast for us is between no nominalization (PNC) and nominalization (control). The question of which head, n or D, is the nominalizer is of secondary interest. For simplicity we will assume only one nominal layer above γP , namely D. As far as we can tell, our analysis is compatible with a double n-D layer structure with n doing the nominalizing. See also Carrier and Compton (2024) for relevant discussion of nominalized *neq*-structures in Inuktitut.

5.1 γ P

In her discussion of control structures, Bittner (1994:53-54,64) analyzes *-neq* as a noun. While this could potentially account for the nominal properties of *neq*-phrases in control structures—the *neq*-phrase bears case, controls agreement, scrambles, and triggers ergative on the subject—it wouldn't extend to *neq*-phrases in the PNC, which have none of these nominal characteristics. Rather than posit two distinct *neq*-heads, we pursue the hypothesis that *-neq* is not a nominalizer, but a functional head on the verbal spine which is encased in a DP-shell in control structures, but not in the PNC. Since there are no articles in Kalaallisut, this structural difference is invisible on the surface. While there are no articles in Kalaallisut, there is a rich inventory of demonstratives some of which have adnominal uses. Here we take the position that adnominal demonstratives are not D heads, but instead project a distinct Demonstrative Phrase, as advocated by e.g. Giusti (1994), Rosen (2003), Guardiano et al. (2012) and Lewis (2014).

5.2 Structure below γ P

Before we document the syntactic material that *neq* can embed we need to say a few words about the clausal structure we assume for Kalaallisut and about the scope of our investigation. Morphology reveals at least the following heads on the verbal spine: i) V, which corresponds to the verbal root and introduces any internal arguments; ii) Caus which introduces a causer argument and demotes the embedded Agent to oblique, iii) Appl which promotes an oblique argument to direct object; iv) the antipassive head ap (Yuan 2018:205ff), which demotes the notional object to an oblique, and v) Voice, which introduces the Agent in active clauses and hosts the passive suffix in passive clauses. We will not be concerned with the relative height of these heads, but simply document that any and all of them can occur inside γ P. (For ordering of the corresponding derivational morphemes see Fortescue (1980).) We also limit ourselves to argument-structure changing morphology and indisputably inflectional morphology, which is mood and agreement. We do this for two reasons. First, there are 300–400 derivational suffixes that occur in Kalaallisut verbs (Fortescue 1980:259) and it would be practically impossible to examine for each of them whether it can occur inside γ P. Second, the suffixes that we choose to look at are exactly the ones that are most informative for our analytic purpose. Argument-changing morphology interacts with case and agreement, which is what we are analyzing. Inflectional morphology is hosted in C and the very outermost morphology of the verb word. As such, it let's us determine whether the upper bound of *-neq* is below the finite CP.

Our documentation relies on the Mirror Principle which states that “Morphological derivations directly reflect syntactic derivations (and vice versa)” (Baker 1985:375). Of particular relevance is its corollary regarding morpheme order and hierarchy of syntactic projections: if morpheme α occurs closer to the root than morpheme β , then the syntactic head responsible for α is lower in the syntactic structure than the syntactic head responsible for β . We will see that any argument-structure changing morphology can occur closer to the root than *-neq*, which indicates that VP, apP, CausP, ApplP and VoiceP can all be projected below γ P. In the PNC this is true in all circumstances, whereas it is modulated by the ban on multiple absolutes in control structures. In neither construction is inflectional morphology inside *-neq* possible.

We start our examination with basic active clauses. As seen above already, these are licit in both the PNC and in control structures.

(59) ACTIVE VOICE

- a. Naja-p meeqqa-t ikior-neq ajor-pai
Naja-ERG child-PL.ABS help-NEQ HAB.NEG-3SG>3PL.IND
'Naja helped the children.' Grethe Schmidt BRI-GS-2
- b. Naja-p mersor-neq nuannari-vaa.
Naja-ERG sew-NEQ love-3SG>3SG.IND
'Naja loves to sew.' Grethe Schmidt CIB-GS-2

γ -phrases formed on passives are also possible in both constructions, with the passive suffix *-neqar* occurring inside *-neq* :

(60) PASSIVE VOICE

- a. Meeqqa-t Naja-mik ikior-**neqar**-neq ajor-put.
child-PL.ABS Naja-INST help-PASS-NEQ HAB.NEG-3PL.IND
'The children are not being helped by Naja.' Ellen Thrane BRN-7
- b. Meeqqa-t Naja-mik ikior-**neqar**-neq nuannari-vaat.
child-PL.ABS Naja-INST help-PASS-NEQ love-3PL>3SG/PL.IND
'The children love to be helped by Naja.' Ellen Thrane CIP-1

The antipassive morpheme *-i* likewise occurs inside *-neq* indicating that apP can also be contained in γ P:

(61) ANTIPASSIVE

- a. Naja meeqqa-nik iku-i-neq ajor-poq.
Naja.ABS child-INST.PL help-AP-NEQ HAB.NEG-3SG.IND
'Naja doesn't help the children.' Ellen Thrane BRP-1
- b. Naja-p meeqqa-nik iku-i-neq nuannari-vaa.
Naja-ERG child-INST help-AP-NEQ love-3PL>3SG.IND
'Naja loves to help the children.' Ellen Thrane CIP-3

The causative suffix *-tip* can occur inside *-neq* in the PNC (62).⁷

- (62) Juuna-p Naja-mut meeqqa-t ikior-**tin**-neq ajor-pai.
Juuna-ERG Naja-ALL child-PL.ABS help-CAUS-NEQ HAB.NEG-3SG>3PL.IND
'Juuna doesn't let Naja help the children.' Ellen Thrane CIK-1

In contrast, causative *-tip* inside *-neq* is ungrammatical in the control example in (63).

- (63) *Juuna-p Naja-mut meeqqa-t ikior-**tin**-neq nuannari-vaa.
Juuna-ERG Naja-ALL child-PL.ABS help-CAUS-NEQ love-3SG>3SG.IND
Intended: 'Juuna loves letting Naja help the children.' Ellen Thrane CFK-11

⁷As discussed by Fortescue (1984:84ff), *-tip* has a reading akin to an analytic 'let'-causative, as in (62). Also note that the final consonant of the causative morpheme assimilates to the initial consonant of *-neq* producing the surface form *-tin-neq*.

We attribute this contrast to the single absolutive restriction: in (62) there is only one absolutive DP, namely the *meeqqa-t* ‘children’. In (63) there are two: *meeqqa-t* and the nominalized DP that contains it. Evidence that it is indeed the single absolutive restriction that is at play and not a difference in whether γ P can embed CausP in the two constructions comes from the example in (64). This is an antipassivized version of (63). As such the theme is not absolutive, but instrumental and the entire construction contains only one absolutive, namely the DP complement of the control verb. The grammaticality of (64) thus confirms that γ can embed CausP in a control construction, just as it can in the PNC.

- (64) Juuna-p Naja-mut meeqqa-nik ikiu-i-**tin-neq** nuannari-vaa.
 Juuna-ERG Naja-ALL child-PL.INS help-AP-CAUS-NEQ love-3SG>3SG.IND
 ‘Juuna loves letting Naja help the children.’ Ellen Thrane CKP-1

The applicative morpheme *-ute* has multiple functions (Woodbury 1977), two of which are relevant here. First, it transitivity an intransitive verb (65). Second, when applied to a ditransitive verb, it promotes an oblique theme to direct object, as seen in (66b). (The underived form in (66a) shows that Kalallisu has secundative alignment in basic ditransitive structures.)

(65) TRANSITIVIZING FUNCTION OF -UTE

- a. Ummi-vunga.
 aim-1SG.IND
 ‘I took aim.’ Ellen Thrane CKQ-1
- b. Orpi-it ummi-**up**-pakka.
 tree-PL.ABS aim-APPL-1SG>3SG.IND
 ‘I aimed for the trees.’ Ellen Thrane CKQ-2

(66) THEME PROMOTING FUNCTION OF -UTE

- a. Meeqqa-t poortukka-nik nassip-pakka.
 child-PL.ABS package-PL.INST send-1SG>3PL.IND
 I sent the children the packages. Ellen Thrane CIM-1
- b. Meeqqa-nut poortukka-t nassi-**up**-pakka.
 child-PL.ALL package-PL.ABS send-APPL-1SG>3PL.IND
 I sent the packages to the children. Ellen Thrane CIM-2

The PNC allows both applicative structures to surface inside the *neq*-phrase:⁸

- (67) a. Orpi-it ummi-**un-neq** ajor-pakka.
 tree-PL.ABS aim-APPL-NEQ HAB.NEG-3SG>3PL.IND
 ‘I don’t aim for the trees.’ Ellen Thrane CKQ-4
- b. Meeqqa-nut poortukka-t nassi-**un-neq** ajor-pakka.
 child-PL.ALL package-PL.ABS send-APPL-NEQ HAB.NEG-1SG>3PL.IND
 ‘I didn’t send the package to the children.’ Grethe Schmidt CJM-GS-1

⁸By assimilation to the following /n/, *-ute* surfaces as *-un*.

In the control structure, we again see the effect of the single absolutive restriction. Specifically, the structures in (68) are ungrammatical because, in addition to the absolutive *neq*-DP, there is a second absolutive DP (*orpiit* in (68a) and *poortukkat* in (68b)).

- (68) a. *Orpi-it ummi-**un-neq** qatsup-para.
 tree-PL.ABS aim-APPL-NEQ be.tired.of-1SG>3SG.IND
 Intended: ‘I am tired of aiming for the trees.’ Ellen Thrane CKR-2
- b. *Meeqqa-nut poortukka-t nassi-**un-neq** nuannari-vara.
 child-PL.ALL package-PL.ABS send-APPL-NEQ love-1SG>3SG.IND.
 Intended: ‘I love to send packages to the children.’ Ellen Thrane CKI-3

Again we can show that the issue is not embedding ApplP under γ , but instead the ban against multiple absolutives. Starting with the ungrammatical applicative structure in (68a), we can avoid multiple absolutives by antipassivizing the verb so that the theme is realized with instrumental case. This strategy is confirmed by the grammaticality of (69a). Similarly, we can antipassivize the ditransitive verb in (68b) to produce the grammatical (69b); the final /p/ of the causative suffix assimilates to the initial /s/ of the antipassive morpheme.

- (69) a. Orpin-nik ummi-us-si-neq qatsup-para.
 tree-PL.INST aim-APPL-AP-NEQ be.tired.of-1SG>3SG.IND
 ‘I am tired of aiming for the trees.’ Ellen Thrane CKR-8
- b. Meeqqa-nut poortukka-nik nassi-us-si-neq nuannari-vara.
 child-PL.ALL package-PL.INST send-APPL-AP-NEQ love-1SG>3SG.IND.
 ‘I love to send packages to the children.’ Ellen Thrane CLO-3

Thus γ can embed active and passive VoiceP, ApplP, CausP and apP. In contrast *neq*-phrases verbs cannot contain inflectional morphology, specifically mood and agreement morphology, in neither the PNC (70) nor control (71).

- (70) *Naja biiler-**luni**-neq ajor-poq.
 Naja.ABS drive-4SG.CONT-NEQ HAB.NEG-3sg.ind
 Intended: ‘Naja doesn’t drive.’ Ellen Thrane CIR-2
- (71) *Naja-p biiler-**luni**-neq nuannari-vaa.
 Naja-ERG drive-4SG.CONT-NEQ.ABS love-1SG>3SG.IND
 Intended: ‘Naja loves to drive.’ Ellen Thrane CIR-4

These examples are modeled on mood and agreement morphology in finite embedded clauses. When the subject of a finite embedded clause is co-referential with the matrix subject, contemporative mood (72) is used in place of the usual participial mood (73):

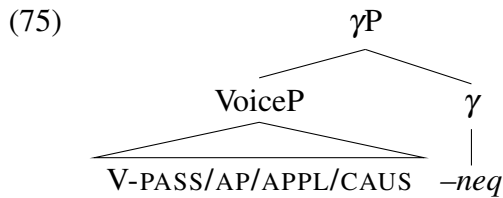
- (72) Naja oqar-poq tamatigut angerla-jaar-luni.
 Naja.ABS say-3SG always leave-early-4SG.CONT
 Naja_i said that she_i always leaves early. Ellen Thrane BCC-2
- (73) Naja oqar-poq illin=una angerla-jaar-tutit.
 Naja.ABS say-3SG 2SG=FOC leave-early-2SG.PART
 Naja said that it was you that left early. Ellen Thrane BDX-4

In (70) and (71) the embedded subject is similarly co-referential with the matrix subject and we therefore expect contemporative mood. However, it is ungrammatical to affix the 4SG contemporative suffix *-luni* inside of *neq*. The data provided in this section are summarized in the table below. Recall that for the causative and applicatives, the control construction requires antipassivization to not run afoul of the single absolutive restriction.

(74) POSSIBLE STRUCTURE BELOW -NEQ

STRUCTURE	PNC	CONTROL
active	✓	✓
passive	✓	✓
antipassive	✓	✓
applicative of intransitive	✓	✓
applicative of transitive	✓	✓
causative	✓	✓
agreement/mood	✗	✗

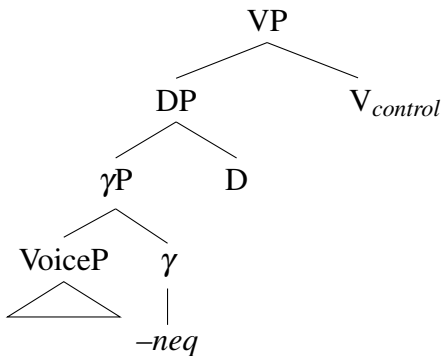
On our structural analysis of γP (repeated below), this distribution is due to γ invariably taking VoiceP as its complement and never CP, as would be required for mood and agreement morphology to occur inside *-neq*.



5.3 Structure above γP

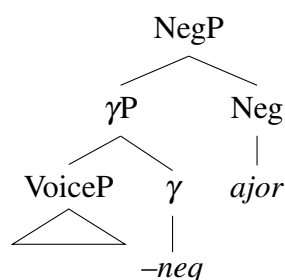
In control structures γP is encased in a DP shell, which in turn occupies the complement position of the control verb.

(76) CONTROL



In the PNC, γP is not encased in a DP shell but is the complement of the negative auxiliary. As for the negative auxiliary itself, we propose that it projects a NegP as in Holmberg et al.'s (1993:183–185) analysis of negative auxiliaries in Finnish.

(77) PNC



Evidence for a NegP analysis comes from the incompatibility of the negative auxiliary and suffixal negation *-nngit*:⁹

- (78) *Naja biiler-neq **aju-nngi**-laq.
 Naja.ABS drive.car-NEQ HAB.NEG-neg-3SG.IND.NEG
 Intended: ‘Naja drives. [lit. Naja doesn’t not drive]’

Grethe Schmidt BSR-GS-4

We note that the lexical verb *ajor-* ‘be bad’ can co-occur with suffixal negation as in the example in (79).

- (79) pisiari-gukku **aju-ssa-nngi**-laq
 buy-2SG>3SG.COND be.bad-FUT-NEG-3SG.IND.NEG
 ‘It would be good if you bought it.’

Fortescue 1984:33

In accord with the ungrammaticality judgment in (78), we have not found any examples of negative auxiliary *ajor* and *-nngit* cooccurring in Bittner’s text collection or in Fortescue (1984), Sadock (2003), and Kahn and Valijärvi (2022). In addition to being in complementary distribution with *-nngit*, both morphemes are followed by mood and agreement morphology as seen in (80).

- (80) a. Bilii-**nngi**-langu.
 drive-NEG-1SG.IND.NEG
 ‘I don’t drive.’
 b. biiler-neq **ajor-punga**.
 drive-NEQ HAB.NEG-1SG.IND
 ‘I don’t drive.’

Grethe Schmidt CMB-GS-2

Grethe Schmidt CMB-GS-3

Moreover, in its habitual negation use, *ajor-* does not stand in a thematic relationship to any of the DPs it co-occurs with—i.e. *ajor-* does not assign any theta roles—suggesting that it is not a lexical verb. Finally, the fact that *ajor-* is a *negative* auxiliary makes it a good fit for Neg⁰. Our conclusion is therefore that *ajor-* projects a NegP, as illustrated in (77).

⁹There are three morpho-phonological processes at play in (79). First the negative suffix deletes the final /t/ of the negative auxiliary. This in turn bleeds the phonological rule that lowers /u/ to /o/ before /t/. Third *-nngit* triggers a special form of the 3SG.IND morpheme, *-laq* in place of the regular form *-poq*.

6 Implications for case

Our examination of the PNC and control constructions affirms the following generalization about case in Inuit. (This is also summarized in (Compton 2018:164)):

(81) SINGLE ABSOLUTIVE RESTRICTIONS (preliminary)

There is at most one absolutive DP per clause.

The Single Absolutive Restriction is motivated by the lack of ditransitives with two absolutive internal arguments, as in (82).

(82) *Meeqqa-t poortukka-t nassip-pakka.
child-PL.ABS package-PL.ABS send-1SG>3PL.IND

Intended: ‘I sent the children the packages.’

Ellen Thrane CKS-2

We have shown above that it extends to control structures with two absolutive arguments: the nominalized *neq*-DP and an absolutive DP within it. PNCs with only an absolutive argument of the lexical verb don’t run afoul of the Single Absolutive Restriction since, in the PNC, γP is not encased in a D-shell and thus an absolutive argument of the lexical verb is the sole absolutive argument in this structure.

The non-finite constructions considered above further allows us to refine the domain in which the Single Absolutive Restriction holds. In particular, we argue for the more specific version of the Single Absolutive Restriction in (83), that references case domains.

(83) SINGLE ABSOLUTIVE RESTRICTION (final)

There is at most one absolutive DP per case domain, where

- a. finite CP is a case domain
- b. nothing else is a case domain

Let us first establish that this formulation of the Single Absolutive Restriction accounts for the ungrammaticality of two absolutive internal arguments in a ditransitive clause:

(84) *Meeqqa-t poortukka-t nassip-pakka.
child-PL.ABS package-PL.ABS send-1SG>3PL.IND

Intended: ‘I sent the children the packages.’

Ellen Thrane CKS-2

In (84) there only one case domain, namely the finite CP, but two absolutive DPs. This state of affairs is straightforwardly ruled out by (83). Turning to control structures, the relevant configuration is one with a transitive embedded verb, as in (85):

(85) a. *Iipili-t neri-neq nuannari-vara.
apple-PL.ABS eat-NEQ love-1SG>3SG.IND

Intended: ‘I love to eat apples.’

Ellen Thrane CKF-3

- b. $[_{CP} [_{DP_2} DP_1 [_{\gamma P} V\text{-}neq]] V_{fin}]$

In 85 there is also only one case domain. While there are two clauses, only one of them is a finite CP, namely the higher clause; the lower clause is just a γ P. Since there is exactly one case domain, there should be at most one absolutive by (83), but there are two. One is the internal argument of the embedded verb, DP₁, the second is the nominalized *neq*-phrase, DP₂.

This example motivates the restriction on case domains to finite CPs and not just any verbal projection. If γ P formed its own case domain, (85) would be predicted to be grammatical. There would be two case domains: γ P and the higher CP. More precisely the higher CP case domain would be the matrix clause, excluding γ P. Inside the γ P case domain there is exactly one absolutive DP, DP₁. Inside the CP case domain there is also exactly one absolutive DP, namely DP₂. Similarly, if DP were a case domain, there would be one absolutive per case domain in (42b) and it should be grammatical contrary to fact.

The claim that only a finite CP constitutes a case domain in Kalaallisut can be tested in at least two additional environments: i) finite clausal complements and ii) affixal verb constructions.

Finite clausal complements In addition to control structures that embed nonfinite clauses, Kalaallisut has a large inventory of verbs that embed finite clauses (Fortescue 1984:34–44), including *aperi*- ‘ask’ (86) and *nassuiar*- ‘explain’ (87).

- (86) *Meeqqa-t aperi-vakka [illit angerla-jaar-ner-sutit].*
 child-PL.ABS ask-1SG>3PL.IND 2SG.ABS leave-early-if-2SG.PART
 ‘I asked the children if you left early.’ Ellen Thrane AIA-1

- (87) *Naja nassuiar-para [illit poortukka-t Juuna-mut*
Naja.ABS explain-1SG>3SG.IND 2SG.ERG package-PL.ABS Juuna-ALL
nassi-uk-kitit].
send-APPL-2SG>3PL.PART
 ‘I explained to Naja that you sent the packages to Juuna.’ Ellen Thrane BET-1

Of importance here is that each of these examples contain two absolutives; the indirect object in the matrix clause and an intransitive subject or transitive object in the embedded clause. Unlike the control example in (85a), there are two finite CPs in (86) and (87) and therefore two case domains with exactly one absolutive DP in each, in accordance with the Single Absolutive Restriction in (83).

Incorporation In addition to noun incorporation, Kalaallisut has a number of verb roots that incorporate a VP (Fortescue 1984:324ff). This VP incorporation structure is illustrated in (88) with the incorporating root *-qqu* ‘ask’.

- (88) *Naja-p Hansi angerla-qqu-aa.*
Naja-ERG Hansi.ABS come.home-ask-3SG>3SG.IND
 ‘Naja asked Hansi to come home.’ Ellen Thrane CKT-1

Notice that the subject of the incorporated verb, here *Hansi*, is realized as an absolutive DP. If a transitive verb is incorporated there are thus potentially two absolutives in the structure, namely the subject and the object of the incorporated verb. As shown in (89) such double-absolutive structures are ungrammatical. Instead three strategies that eliminate a second absolutive can be employed.

The first is for the embedded Agent to be realized in the allative case (90). The second is for the embedded theme to be realized in the instrumental, taking advantage of *neri-* being half-transitive (91). Finally, the theme can be incorporated as in (92).

- (89) *Naja-p quaq Hansi neri-qqu-aa.
 Naja-ERG frozen.meat.ABS Hansi.ABS eat-tell-3SG>3SG.IND
 Intended: ‘Naja told Hansi to eat frozen meat.’ Ellen Thrane CKT-2
- (90) Naja-p quaq Hansi-mut neri-qqu-aa.
 Naja-ERG frozen.meat.ABS Hansi-ALL eat-ask-3SG>3SG.IND
 ‘Naja told Hansi to eat frozen meat.’ Ellen Thrane CKT-5
- (91) Naja-p Hansi qua-mik neri-qqu-aa.
 Naja-ERG Hansi.ABS frozen.meat-INST eat-ask-3SG>3SG.IND
 ‘Naja told Hansi to eat frozen meat.’ Ellen Thrane CKT-4
- (92) Naja-p Hansi quar-to-qqu-aa.
 Naja-ERG Hansi.ABS frozen.meat-consume-ask-3SG>3SG.IND
 ‘Naja asked Hansi to eat frozen meat.’ Ellen Thrane CKT-3

The VP headed by *neri-* and embedded under *-qqu* is non-finite and hence does not constitute its own case domain. Instead it forms a single case domain with the matrix clause headed by *-qqu*. As such (83) requires that there be at most one absolutive in the entire sentences, which is true for the grammatical examples in (90)–(92) but not for the ungrammatical (89).

Exactly vs. at most one absolutive per case domain In all the constructions discussed so far, there is exactly one absolutive per case domain. This raises the possibility that the Single Absolutive Restriction could be further refined as in (93).

- (93) SINGLE ABSOLUTIVE RESTRICTION (to be rejected)
 There is exactly one absolutive per case domain.

To evaluate (93) against (83) we turn to two existential constructions. The first is the participial incorporation construction in (94).¹⁰

- (94) Arna-nik pingasu-nik tikit-to-qar-poq.
 woman-PL.INST three-PL.INST arrive-3SG.PART-have-3SG.IND
 ‘There arrived three women.’ Ellen Thrane CLP-2

Here the lexical verb is in the participial form and incorporated into the root *-qar*. As indicated by the final mood and agreement suffix *-poq* this construction involves a finite CP, i.e. a case domain, and yet there appears to be no absolutive DP, only the instrumental DP *arnanik pingasunik* ‘three women’. This is compatible with our version of Single Absolutive Restriction which says that there is at most one absolutive per case domain (83), but violates the hypothetical stronger Single Absolutive Restriction in (93). Additional support for (83) comes from an impersonal passive construction reported in Bittner and Hale (1996b) and illustrated in (95).¹¹

¹⁰This construction is also found in other Inuit languages, suggesting that (93) is too strong also in these varieties. See Yuan (2022b) for detailed discussion of the construction in Inuktitut.

¹¹This construction is not part of the grammars of the Third Author and Fourth Author.

- (95) Ippassaq qitin–niqar–poq.
yesterday dance–PASS–3SG.IND

‘There was some dancing yesterday.’

Bittner and Hale (1996b:32, ex (50a))

In (95) the intransitive verb *qitik* ‘dance’ is passivized. As it is an unergative verb, there is no patient DP and also no expression of the Agent. Taking (95) at face value, this is another well-formed finite CP without an absolutive argument. We thus conclude that the Single Absolutive Restriction should be stated as a ban on multiple absolutives in a single case domain, and not as a requirement that every finite CP has exactly one absolutive argument.

Absolutives and finiteness Above we have argued that only finite clauses license an absolutive DP, non-finite clauses do not. This makes Kalaallisut a HIGH-ABS language in the typology of Coon et al. (2014). In a HIGH-ABS language, absolutive DPs are licensed by Infl. Infl is absent in non-finite clauses, which therefore do not license absolutive arguments. This categorization connects to a restriction on A-bar extraction in the Kalaallisut: ergative arguments cannot be relativized, only absolutive arguments can (Fortescue 1984:50, Bittner 1994:58). Coon et al. (2014) argue that this Ergative Extraction Constraint (Aissen 2017) is only active in high-abs languages, which fits with the facts of Kalaallisut.¹²

Non-verbal clauses Before we leave this section, we need to say something about non-verbal clauses, since they appear to violate the Single Absolutive Restriction, both in its original and refined form. There are several ways of forming non-verbal clauses in Kalaallisut (Mikkelsen and Thrane 2024). One of them is illustrated in (96), where the particle *tassa* is flanked by two absolutive DPs (Fortescue 1984:72ff). The relevance of (96) for the Single Absolutive Restriction is immediately clear: there are two absolutive DPs in the clause and yet it is grammatical. Motivated by its lack of agreement and mood morphology, we propose that (96) is not a CP, but a bare equative small clause headed by *tassa* (on equative small clauses see e.g. Heycock and Kroch 1999.) As such, it does not form a case domain and (83) does not apply.¹³

- (96) Pisortaq tassa Hansi.
boss.ABS TASSA Hansi.ABS

‘The boss is Hansi.’

Ellen Thrane AVA-1

¹²Legate (2008) draws a distinction between ABS=NOM and ABS=DEF. In a ABS=NOM system, absolutive case is licensed by Infl. In a ABS=DEF system absolutive is a morphological default that spells out both nominative and accusative syntactic case in the absence of dedicated exponents for each of these. Under the HIGH-ABS analysis outlined above, Kalaallisut is a ABS=NOM language in Legate’s term. Interestingly, the opposite conclusion—Kalaallisut absolutive is the syncretic realization of nominative and accusative case—is reached by Deal et al. (2024), based on the nominative-accusative alignment of certain quantifiers and focus markers. While we cannot resolve this tension here, we note that it could motivate a revision of Legate’s classification.

¹³Two other interpretations of (96) are possible, both compatible with the Single Absolutive Restriction. The first interpretation is that the DPs flanking *tassa* are actually caseless. Under that interpretation, (96) is in accord with (83), whether (96) constitutes a case domain or not. A second interpretation of (96) is articulated in Mikkelsen and Thrane (2024:14-15). On their view only *pisortaq* is an argument and *Hansi* is a non-verbal predicate that is not case-marked. Again, this interpretation is compatible with the Single Absolutive Restriction whether or not (96) forms a case domain.

A second strategy for non-verbal predication is noun incorporation into the copular root *-u*. In (97), the predicate nominal *siku* ‘ice’ is incorporated into the copular root *-u* and the resulting verb predicates over the subject *imaq* ‘sea’.

(97) For during that month [April] anywhere

imaq siku-u-mi-sar-poq
 sea.ABS ice-COP-partly-HAB-3SG.IND

‘the sea is [at least] partly frozen’ [lit. The sea is partly ice.]

which is very good for those who intend to travel.

Qillarsuakkunik oqalualaaq (Qillarsuaq’s Saga) Passage 1: Flight from Canada, §5, ex 4

Since incorporated nouns generally do not bear case (Sadock 1980/84), (97) does not present a challenge to the Single Absolutive Restriction. There is exactly one absolutive in the clause, namely *imaq*. However, stranded modifiers of incorporated predicate nominals surface in the bare form as seen in (98).¹⁴

(98) *Ivalu politikeri-u-voq ilumoortoq.*
Ivalu.ABS politician-COP-3SG.IND honest

‘Ivalu is an honest politician’

Ellen Thrane BHL-4

On the face of it, this looks like a single finite CP containing two absolutives in violation of (83). One possible resolution is that there are not actually two DPs in (98). The subject *Ivalu* is clearly a DP, but perhaps the modifier *ilumoortoq* ‘honest’ is simply a morphologically bare AP not contained in a DP shell. Since the Single Absolutive Restriction regulates only DPs, (98) would therefore not actually violate it.

7 Behavior of other negative auxiliaries

At the beginning of the paper we introduced *ajor-* ‘HAB.NEG’, *artor-* ‘be unable’, *nalu* ‘not know’, and *saper-* ‘not manage’ as a single class of “negative auxiliaries”, in line with existing descriptions. In this section we show that there are individual differences between these five verbs and that, in fact, *ajor-* is unique among them.

saper- ‘not manage’ This verb differs from *ajor-* in that it participates in both the PNC (99) and the control construction (100):

(99) *Naja mersor-neq saper-poq.*
Naja.ABS sew-NEQ not.manage-3SG.IND
 ‘Naja can’t sew.’

Ellen Thrane CIT-1

¹⁴This sets copular incorporation apart for incorporation into other roots, where stranded modifiers surface with instrumental case morphology. See Mikkelsen and Thrane (2024) for detailed discussion.

- (100) Naja-p mersor-neq saper-paa.
 Naja-ERG sew-NEQ not.manage-3SG>3SG.IND
 ‘Naja can’t sew.’

Ellen Thrane CLP-4

Both examples feature an intransitive embedded verb, but differ in case and agreement. In (99), the subject is absolutive and *saper-* shows intransitive agreement, both characteristic of the PNC. In (100), the subject is ergative and *saper-* agrees with both the subject and the nominalized *neq*-phrase, both characteristic of control constructions.

nalu- ‘not know’ *nalu-* likewise participates in both the PNC (101) and control (102).

- (101) Naja mersor-neq nalu-voq.
 Naja.ABS sew-NEQ not.know-3SG.IND
 ‘Naja doesn’t know how to sew.’
- (102) Naja-p mersor-neq nalu-vaa.
 Naja-ERG sew-NEQ not.know-3SG>3SG.IND
 ‘Naja doesn’t know how to sew.’

Ellen Thrane CIT-3

Ellen Thrane CKX-1

The PNC example in (101) shows intransitive agreement and an absolutive subject, whereas the control example in (102) shows transitive agreement and an ergative subject.

artor- ‘be unable’ This verb differs from both *saper-/nalu-* and *ajor-* in that it only participates in control and is ungrammatical in the PNC frame.

- (103) *Naja mersor-neq artor-poq.
 Naja.ABS sew-NEQ be.unable-3SG.IND
 Intended: ‘Naja is unable to sew.’
- (104) Naja-p mersor-neq artor-paa.
 Naja-ERG sew-NEQ be.unable-3SG>3SG.IND
 ‘Naja is unable to sew.’

Ellen Thrane CIT-2

Elleb Thrane CKX-2

ajornar- ‘be impossible’ This verb is different yet again in that it always shows intransitive agreement, even when the *neq*-marked verb is transitive, as in (105) and (106).

- (105) Karsi kivin-neq ajornar-poq.
 box.ABS lift-NEQ be.impossible-3SG.IND
 ‘The box cannot be lifted.’
- (106) *Karsi kivin-neq ajornar-paa.
 box.ABS lift-NEQ be.impossible-3SG>3SG.IND
 Intended: ‘The box cannot be lifted.’

Grethe Schmidt CJG-GS-8

Ellen Thrane CKV-1

We propose that *ajornar-* involves *tough*-movement: the embedded object moves out of the non-finite VP to the matrix subject position, cf. the English example in (107).

- (107) The box_i is impossible [to lift ___i].

The *neq*-phrase is a γ P and hence not case marked. As the sole absolutive argument in a finite CP *karsi* ‘box’ meets the Single Absolutive Restriction. In the *tough*-movement analysis *karsi* is the surface subject of *ajornar*-. It therefore predicts that a plural internal argument should trigger plural agreement on the negative auxiliary verb. This prediction is borne out by the examples in (108) and (109):

- (108) *Karsi-t kivin-neq ajornar-poq.
 box-PL.ABS lift-NEQ be.impossible-3SG.IND
 Intended: ‘The boxes cannot be lifted.’ Ellen Thrane CKV-3
- (109) Karsi-t kivin-neq ajornar-put.
 box-PL.ABS lift-NEQ be.impossible-3PL.IND
 ‘The boxes cannot be lifted.’ Ellen Thrane CKV-4

The behavior of the negative verbs discussed so far in this paper is summarized in the table in (110).

(110)

	PNC	CONTROL
<i>ajor</i> - ‘not’	✓	✗
<i>saper</i> - ‘not manage’	✓	✓
<i>nalu</i> - ‘not know’	✓	✓
<i>artor</i> - ‘be unable’	✗	✓
<i>ajornar</i> - ‘be impossible’	✗	✗

A number of conclusions can be inferred from this table. First of all, the five verbs identified as “negative auxiliaries” in descriptive grammars do not show uniform behavior. *ajor*- is unique in participating only in the PNC. Despite being identified as a negative auxiliary, *artor*- cannot be used with PNC syntax, only in control constructions. In other words, *artor*- is a control verb that happens to have a negative meaning. *saper* and *nalu*- both participate in both PNC and control and thus form a mini class on their own. Finally *ajornar*- has a different syntax altogether. It does not show PNC syntax or control behavior, but instead seems to be a *tough*-construction.

8 Conclusion and open issues

In this paper we have described and analyzed two Kalaallisut constructions involving a non-finite verb formed with the suffix *-neq*: perihastic negation and control. We showed that the two constructions differ systematically in case and agreement properties and argued that this follows from the *neq*-phrase being nominalized in control structures but not in the PNC. This analysis sheds new light on the nature of case domains in the language and the distribution of absolutives. We demonstrated that the case domain is the finite CP and that a case domain can contain at most one absolutive DP.

In previous work, a number of negative verbs have been lumped together as a class, but we showed that there are individual differences and classified each verb as PNC only, control only, both or neither. A next step is to try to understand why each verb behaves the way it does. We suspect that there might be a semantic component to such an explanation. The fact that two negative

verbs (*saper-* ‘not manage’ and *nalu-* ‘not know’) participate in both PNCs and control constructions suggests that PNC verbs diachronically evolve from control verbs. Whereas control verbs licenses thematic arguments, their functionalization (semantic reduction) in PNCs entails a loss of this licensing ability leading to syntactic functional restructuring (Cinque (2001), Wurmbrand (2004)). In this view, the use of *ajor-* as a negative habitual auxiliary is related to a transitive form that is semantically similar to *saper-* found in the Kalaallisut–Danish dictionary *Oqaasit*:

(111) *ajor-paa*: ‘can’t manage to, doesn’t have the power to’

The fact that the third and fourth coauthors reject transitive *ajor-* suggests that in their grammar the syntactic change from control verb to negative auxiliary is complete.

This study is important from a Kalaallisut-internal perspective, because it examines periphrasis within a deeply polysynthetic language. Our study is also important within Inuit, since the PNC has not been reported for other Inuit languages. With a more detailed description of the Kalaallisut PNC at hand, it might be possible to determine whether analogous constructions do or do not exist in other Inuit varieties.

Finally, our study makes a typological contribution. We have established that Kalaallisut has (at least) three negative auxiliaries: *ajor*, *saper-*, and *nalu-*, in addition to the affixal negation *-nngit*. Negative auxiliaries form a rare negation strategy. In the WALS sample only 47 of 1157 languages, or 4%, use negative auxiliaries. As noted above Kalaallisut also have affixal negation. This puts Kalaallisut with only 21 languages, or 2% of the WALS sample, that have multiple negation strategies.¹⁵

We leave a number of questions for future research. First, there is evidence of a second control construction in which the control verb agrees with the internal argument of the embedded verb.

(112) *Iipili-t neri-neq nuannari-vakka.*
apple-PL.ABS eat-NEQ love-1SG>3PL.IND
 ‘I love to eat apples.’

Ellen Thrane CKF-2

If *neq*-phrases are always nominalized in control structures, it is not obvious how to reconcile (112) with the Single Absolutive Restriction.

Second, in their influential analysis of polysynthesis in Inuit, Compton and Pittman (2010) propose that wordhood is determined by phasehood and that the only phases are CPs and DPs. This is at odds with our analysis of the PNC in which γP is spelled out as a separate verb word, despite being neither a CP or a DP. The solution, we propose, is to add γ as a phase head. That would explain why *neq*-phrases are always spelled out as words on their own, even when not encased on a DP shell.

Finally, the Single Absolutive Restriction needs to be examined in the context of existing theories of case in Inuit, including Bittner (1994), Bittner and Hale (1996a, 1996b), Bobaljik (1993), Carrier and Compton (2024), Murasugi (1992), Spreng (2006, 2012) and Yuan (2018, 2022a). This literature mostly focuses on case in finite clauses, though Bittner (1994) and Carrier and Compton (2024) recognize the relevance of nominalized *-neq* constructions. As far as we can tell, the PNC and non-verbal clauses formed with *tassa* have not been considered anywhere in the theoretical literature on Inuit case. We hope to have shown that these constructions have much to offer the theory of case.

¹⁵More work is needed to determine which of these 21 languages have the specific combination of negative auxiliary and negative affix.

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